Saatwik Sairaam Vasamsetti

Full-Stack Web Developer · Software Engineer

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PROFESSIONAL SUMMARY

Full-stack engineer who ships complete systems and proves they work. Built Trade Research, where every number is computed and machine-checked back to its source — leakage-free walk-forward back-testing of six scikit-learn models, conformal prediction intervals, and a FastAPI/PostgreSQL/React execution platform — and reported the honest negative result. Also sole author of an emergency-response platform whose 64-route TypeScript API alone passes 636 assertions across six suites, 74 of them security attacks. Nine certifications across AWS, Google and SAP. Available immediately for software-engineering and data internships.

EDUCATION

VIT-AP University, Amaravati — *2024 – 2029*
Integrated M.Tech — Computer Science & Engineering (in collaboration with Virtusa) — *CGPA: 7.55 / 10*

SR Junior College — Intermediate (Class XII, MPC) — *2022 – 2024 · 94.8%*

DAV Public School — Secondary School (Class X) — *81.17%*

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Kotlin, SQL, Java, HTML, CSS
Backend & Data Stores: Node.js, Fastify, Express, FastAPI, REST design, JWT/RBAC auth, PostgreSQL 16 + PostGIS, MySQL, Drizzle, schema & index design, transactional concurrency
Frontend: React 19, Next.js 16, Angular, Tailwind CSS, Jetpack Compose, progressive web apps, dependency-free canvas UI, accessibility
Data, ML & BI: pandas, NumPy, scikit-learn, Ultralytics YOLO11, ONNX, EDA, walk-forward back-testing, conformal prediction, Monte Carlo, Power BI, Tableau, Matplotlib, Excel dashboards
Cloud, Tooling & QA: AWS, Docker & Compose, Git, GitHub Actions CI, Gradle, Vitest, ESLint

PROJECTS

Trade Research — Verifiable Research & Execution Platform — *Aug – Sep 2026*
Stack: *Python, scikit-learn, pandas, FastAPI, PostgreSQL, React, MetaTrader 5, PyInstaller*
- Built a leakage-free walk-forward back-test of six scikit-learn models against naive and drift baselines, with conformal intervals, Monte Carlo paths and a cost-aware trading simulation.
- Made the numbers checkable: indicators computed in Python, financials from SEC EDGAR XBRL, and a verifier that traces every figure in a generated note back to the data — then reported the honest negative result.

RoadAssist Bharat — Offline-First Emergency Mobility Platform — *Aug – Sep 2026*
Stack: *TypeScript, Fastify, PostgreSQL 16 + PostGIS, Drizzle, Kotlin, Jetpack Compose, Python, YOLO11, Docker, GitHub Actions*
- Sole author of a 64-route Fastify/TypeScript API over a 56-table PostGIS schema, plus a Kotlin Compose Android client and four browser surfaces — 117 commits.
- Verified the API with 636 assertions across six suites, zero failures, including 75 concurrency tests and 74 security attacks that must each be refused.
- Trained four YOLO11 variants on RDD2022 across four countries — best measured mAP50 0.471 on held-out validation, plus a separate nano ONNX tier at ~48 ms/image on CPU for Pi-class hardware.

MachineSense AI — Explainable Industrial Intelligence — *Smart India Hackathon 2026*
Stack: *Python, FastAPI, JavaScript, Canvas*
- Built a FastAPI predictive-maintenance platform — weighted 0-100 health scoring, MAD z-score anomaly detection, remaining-useful-life forecasting and energy optimisation — with every score explained by sensor and a digital twin that simulates a maintenance action first.

OfferPilot — Job-Application Platform for F-1 / OPT Students — *2026*
Stack: *JavaScript, HTML, CSS, localStorage*
- Built a dependency-free static platform — marketing site, onboarding, client dashboard, profile, legal pages and a passcode-gated ops console — with no build step and full localStorage persistence.
- Shipped the RezForge resume-tailoring engine, an interview kanban, a referral programme and analytics charts on a hand-written SVG engine, plus an OPT Runway Calculator computing work-authorisation runway, unemployment-day budget and H-1B lottery timing.
- Labelled the prototype honestly on the page: the match and resume engines are deterministic mocks and no real applications are submitted.

Angular Applications — ShopCart, Notes, Search & Auth — *2026*
Stack: *Angular 22, TypeScript 6, standalone components, signals, Vitest*
- Built four Angular 22 applications on standalone components and signals in a zoneless app: ShopCart, a full e-commerce site; a notes app; a live-filtering search template; and an authentication app with a guarded route.
- ShopCart is the largest: searchable, category-filtered catalogue, product detail pages, a quantity-aware cart with computed totals and accounts behind a route guard, covered by 14 Vitest unit tests across three files.

CERTIFICATIONS & RECOGNITION

- **AWS:** Introduction to AWS · Cloud Practitioner (CLF-C02) Domain 1 · Data Engineer – Associate (DEA-C01) Prep · **Google:** Analytics Certification, to May 2027 (ID 183723043)
- **SAP Learning:** SAP Analytics Cloud · SAP Data Architect · **Forage:** Deloitte, Quantium, GenAI Analytics · Campus Ambassador, SmartED · HackDevengers 1.0